

Evaluating
Emergency Dispatch
Efficiency in a Tiered
EMS System

The EMD Scorecard

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Dispatch drives Fire/EMS Response

- Any response system is only as good as its dispatch
- Some EMD complaints quietly consume resources more than others
- Dispatch decisions affect system availability

How a Tiered Response System Works

- BLS ambulances manage lower-acuity calls
- ALS resources supplement BLS and are not always the primary transport unit
- To match resources to patient needs while preserving ALS availability

Why a Tiered Response?

- Improves system efficiency
- Helps manage staffing limitations
- Helps the system keep up with growing call volume

Blind Spots in Tiered Response



SOME EMD COMPLAINTS CONSUME
RESOURCES MORE THAN OTHERS



THESE EFFECTS ARE LARGELY
INVISIBLE WITHOUT COMPLAINT-
LEVEL ANALYSIS

Challenges in a Tiered Response System

- EMD complaint accuracy (Why I'm here)
- Increased Response Times
- Hidden system strain from upgrades

What is an upgrade?

- An upgrade is when an ALS provider from an ALS Support Piece accompanies a BLS Ambulance during transport to the ER.



Why a Scorecard?



REVEALS WHICH EMD CARDS
TRULY DRIVE SYSTEM IMPACT



QUANTIFIES DOWNSTREAM
OPERATIONAL BURDEN



SUPPORTS DATA-INFORMED
EMD CARD DESIGN AND REVIEW

What the Scorecard Measures

- Upgrade Likelihood (Upgrade Odds)
- Relative Risk (Odds Ratio)
- High Priority Transport (Indicator of Critical Care Need)
- ALS Resource Utilization (System Impact)

▲ Guide Card	Upgrade Odds	Odds Ratio	High Priority	ALS Utilization	▼ Total Score
TRAUMA ASSAULT	3.7	3.0	1.5	1.0	9.2
MOTOR VEHICLE CRASH	3.7	3.0	1.0	1.0	8.7
BLEEDING (NON-TRAUMATIC)	3.0	2.5	1.5	1.0	8.0
ABDOMINAL BACK PAIN	2.5	2.5	1.0	1.0	7.0
FALLS	2.0	2.0	2.0	0.6	6.6
ALLERGIC REACTION	0.5	0.5	1.5	1.0	3.5
BREATHING DIFFICULTY	1.0	1.0	1.5	0.0	3.5
CHEST PAIN	1.0	1.0	1.0	0.0	3.0
SEIZURES	0.0	0.0	2.0	0.4	2.4
STROKE	0.2	0.0	1.5	0.6	2.3

Validation Techniques

- Used statistical measures (odds ratios, confidence intervals, significance testing)
- Tested temporal stability across time periods
- Performed sensitivity analysis on scoring

Background on Prince William County, VA



- Approximately 500K people, 56K calls, 42K of which are EMS
- 22 Stations, 24/7 career with supplemental Volunteers on nights and weekends
- 16 (24/7) transport units (9 ALS, 7 BLS)
- All stations have a Career ALS Support piece, usually an ALS Engine

Before and After Implementation

Dispatch Before Tiered Response

- B
- A
- UNC
- STOP

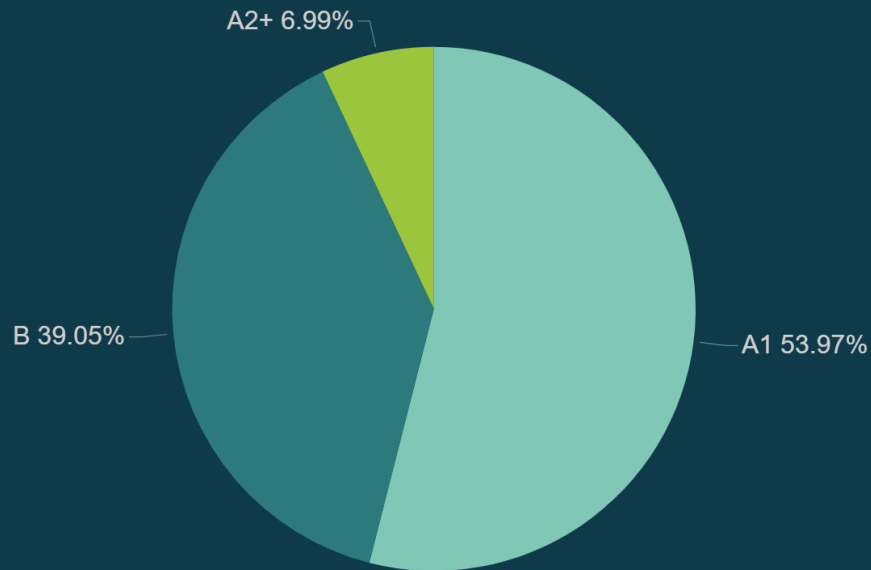
Dispatch After Tiered Response

- B
- A₁
- A₂
- A₃

Dataset Overview: Upgrade Rates

- Overall System: 21.1%
- A1 – 36.1%
- B – 7.4%

Distribution Matters



- System is overwhelmingly BLS Transport driven
- Approximately 93% of EMS calls depend on BLS for initial transport
- Under-triage drives upgrades

57% of A1 Upgrades
came from just 4
Guide Cards

- Seizure
- Stroke
- Chest Pain
- Breathing Difficulty

Dataset Overview: BLS Ambulance Analysis

- Subset of BLS Ambulance Cases Analyzed – 10,609
- A1 – 5,405 cases (52.1%)
- B – 4,972 cases (47.9%)

Possible reasons for so many upgrades

Dispatch

- Lack of data
- Changes made without a clear view of downstream impact

Operational

- Comfort
- Fear of QA
- Patient concern
- Training lag

A₁ became the Default Priority

CHEST PAIN

A₂ Response

Unconscious and/or not breathing normally (agonal)

A₁ Response

Decreased level of consciousness

Weak, dizzy, or faint

Chest pains to include palpitations, discomfort, tightness, or pressure

Pain in jaw, shoulder blades, throat, or teeth

Difficulty breathing / shortness of breath

Chest pain with rapid heart rate or irregular rapid heart rate

Chest pain with nausea / vomiting

Chest pain with sweating / diaphoretic / clammy

Cocaine/crack or other illicit drug use

Prescription drugs such as Viagra, Cialis, Levitra, etc. in the last 48 hours

Pacemaker / AICD (Automatic Cardioverter Defibrillator) activation

B Response

Rapid heart rate without critical symptoms

Chest wall trauma without critical symptoms

Muscle strain

High blood pressure (no critical symptoms)

Why Identification Mattered?



Upgrade Odds: How often was this happening?



Odds Ratio: Which Guide Cards stood out?



High Priority: Were the ALS providers initiating emergent care on Upgrades?



ALS Resource Utilization: How were these upgrades affecting the system?

How often does this Guide Card pull in ALS?

40% - 0-4 Scale

Upgrade Odds

- Measures rate of ALS upgrade by individual guide card.
- Validates each card by comparing its upgrade rate to the system baseline to confirm meaningful differences.
- Example: Motor Vehicle Crashes upgrade infrequently (11.5%) giving them a score of 3.7/4. Strokes upgrade far more (59.6%) for a score of 0.2/4.

Component 1 – Upgrade Odds

Guide Card	Upgrade Rate	Upgrade Odds Score
MOTOR VEHICLE CRASH	11.5%	3.7
BLEEDING (NON-TRAUMATIC)	23.7%	3.0
ABDOMINAL BACK PAIN	25.5%	2.5
FALLS	30.2%	2.0
SICK UNKNOWN	33.3%	2.0
BREATHING DIFFICULTY	40.0%	1.0
CHEST PAIN	41.0%	1.0
DIABETIC	40.9%	1.0
OVERDOSE POISONING	35.8%	1.0
UNCONSCIOUS	41.7%	1.0
STROKE	59.6%	0.2
SEIZURES	71.3%	0.0

Does the Guide Card stand out from others?

30% - 0-3 Scale

Odds Ratio

- Uses logistic regression to measure likelihood of upgrade by individual guide card compared to system average.
- Validates whether a Guide Card's odds differ meaningfully from the system baseline.
- $OR > 1$ = Much more likely than normal. Example: Seizures have more than 4 times higher odds to result in an upgrade than the rest of the system.

Component 2 – Odds Ratio

▲	Guide Card	Odds Ratio	▼	Odds Ratio Score
	CHOKING	0.31		3.0
	MOTOR VEHICLE CRASH	0.23		3.0
	TRAUMA ASSAULT	0.26		3.0
	ABDOMINAL BACK PAIN	0.60		2.5
	BLEEDING	0.55		2.5
	FALLS	0.76		2.0
	BREATHING DIFFICULTY	1.21		1.0
	CHEST PAIN	1.26		1.0
	ALLERGIC REACTION	1.84		0.5
	SEIZURES	4.62		0.0
	STROKE	2.67		0.0

Are ALS providers initiating High Priority Care?

20% - 0-2 Scale

High Priority Care

- Measures whether upgrades resulted in high priority care (Interventions and/or rapid transport).
- Based on the proportion high priority care following the decision to upgrade.
- Example: More frequent rates of high priority care (Chest Pain) produced lower scores further indicating under-triage at dispatch.

Component 3 – High Priority Transport

Guide Card	High Priority Rate	High Priority Score
FALLS	8.8%	2.0
OVERDOSE POISONING	5.3%	2.0
SEIZURES	8.4%	2.0
BREATHING DIFFICULTY	19.8%	1.5
SICK UNKNOWN	15.0%	1.5
STROKE	17.7%	1.5
UNCONSCIOUS	14.6%	1.5
CHEST PAIN	20.3%	1.0
DIABETIC	20.8%	1.0

How do upgrades impact system capacity?

10% - 0-1 Scale

ALS Resource Utilization

- Measures ALS system burden by combining total upgrades and length of time on call per upgrade.
- Converts daily ALS workload into a normalized score to reflect impact on system availability.
- Example: Breathing Difficulty and Seizures generate high daily ALS burden due to frequent, longer upgrades.

Component 4 – ALS Resource Utilization

Guide Card	Daily Burden Index	ALS Resource Utilization Score
ABDOMINAL BACK PAIN	9.3	1.0
BLEEDING (NON-TRAUMATIC)	7.3	1.0
MOTOR VEHICLE CRASH	2.7	1.0
OVERDOSE POISONING	9.7	1.0
DIABETIC	12.0	0.8
FALLS	21.9	0.6
STROKE	21.4	0.6
UNCONSCIOUS	21.9	0.6
SEIZURES	39.0	0.4
BREATHING DIFFICULTY	85.2	0.0
CHEST PAIN	61.7	0.0
SICK UNKNOWN	57.4	0.0

More than just the Scorecard



SCORECARD HELPED IDENTIFY
CHANGES



PREDICTIVE MODELING DROVE
CHANGES USING DEMOGRAPHICS
AND MEDICAL HISTORY FROM
SPECIFIC GUIDE CARDS (LOGISTIC
REGRESSION)



OPERATIONAL MEDICAL
DIRECTOR AND
COMMUNICATIONS SUBJECT
MATTER EXPERTS ASSISTED WITH
THE PROCESS

Upgrade Risk Factors

- Chest Pain calls were 69% more likely to be upgraded if over 50 and 74% more likely to be upgraded with pre-existing cardiac history
- Breathing Difficulty were 73% more likely to be upgraded if over 50 and 41% more likely to be upgraded with pre-existing cardiac history

Revised Dispatch Criteria

CHEST PAIN

A2 Response

Chest pain with history of cardiac events

Pacemaker / AICD (Automatic Implantable Cardioverter Defibrillator) activation

Patient age 50+

A1 Response

Chest pain to include palpitations, discomfort, tightness, or pressure

Chest pain with any drug use

Chest pain with nausea, vomiting, diaphoresis, or clamminess

Chest pain with rapid heart rate or irregular heart rate

Decreased level of consciousness

Difficulty breathing / shortness of breath

Low blood pressure

Pain in jaw, shoulder blades, throat, or teeth

B Response

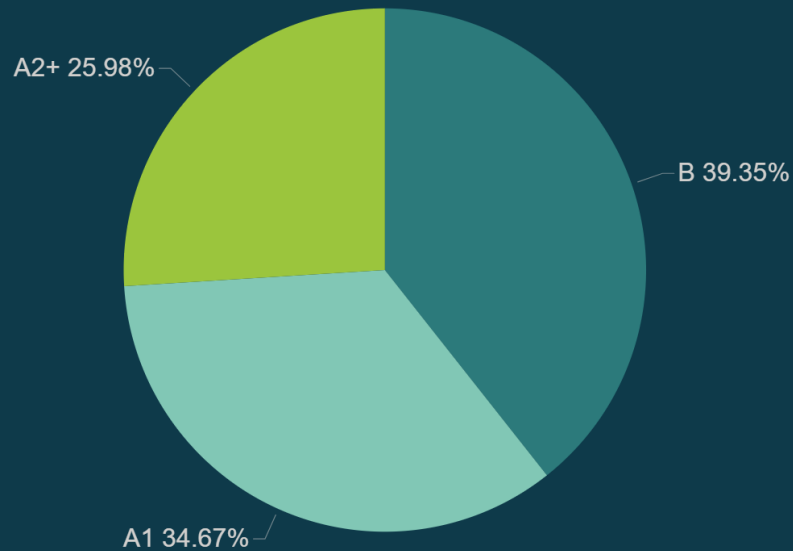
Chest Wall Trauma (no critical criteria)

High blood pressure (no critical criteria)

Muscle strain

Rapid heart rate / anxiety / panic attack (no critical criteria)

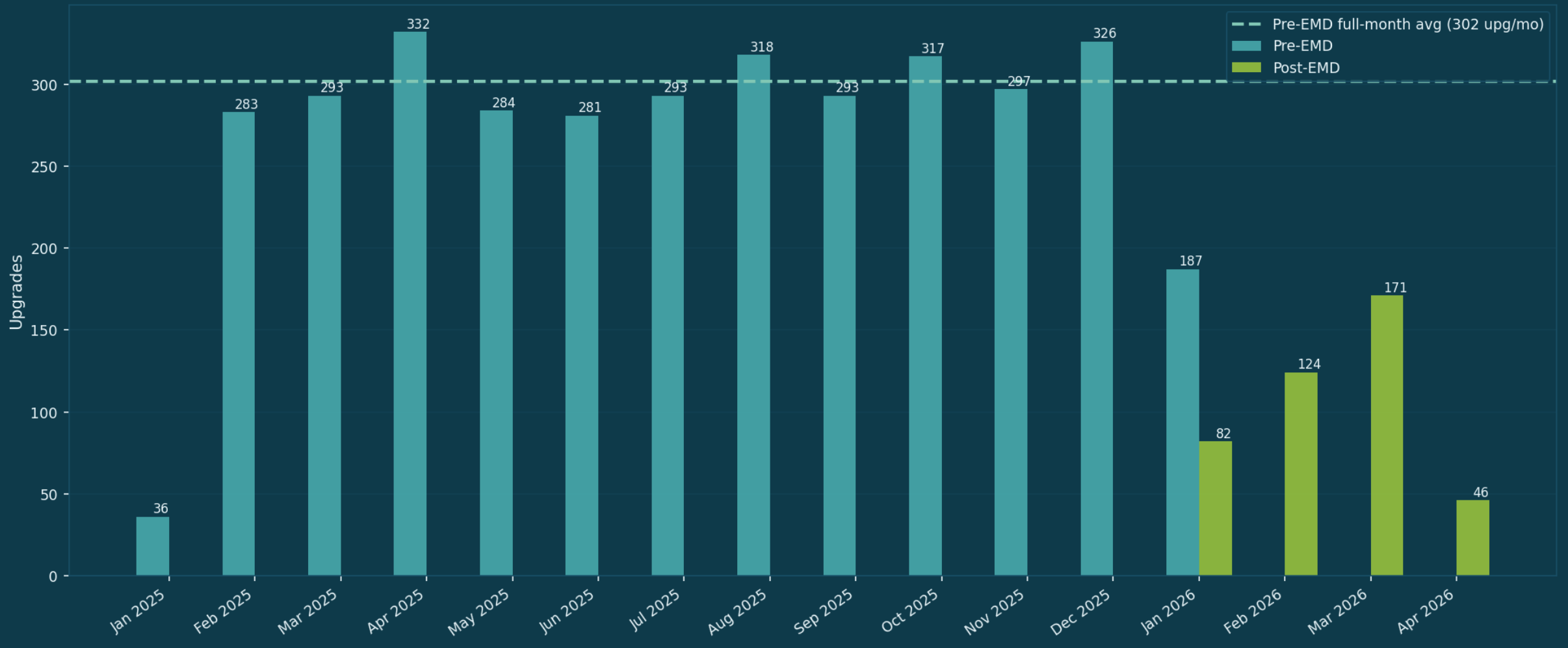
Where Are We Now?



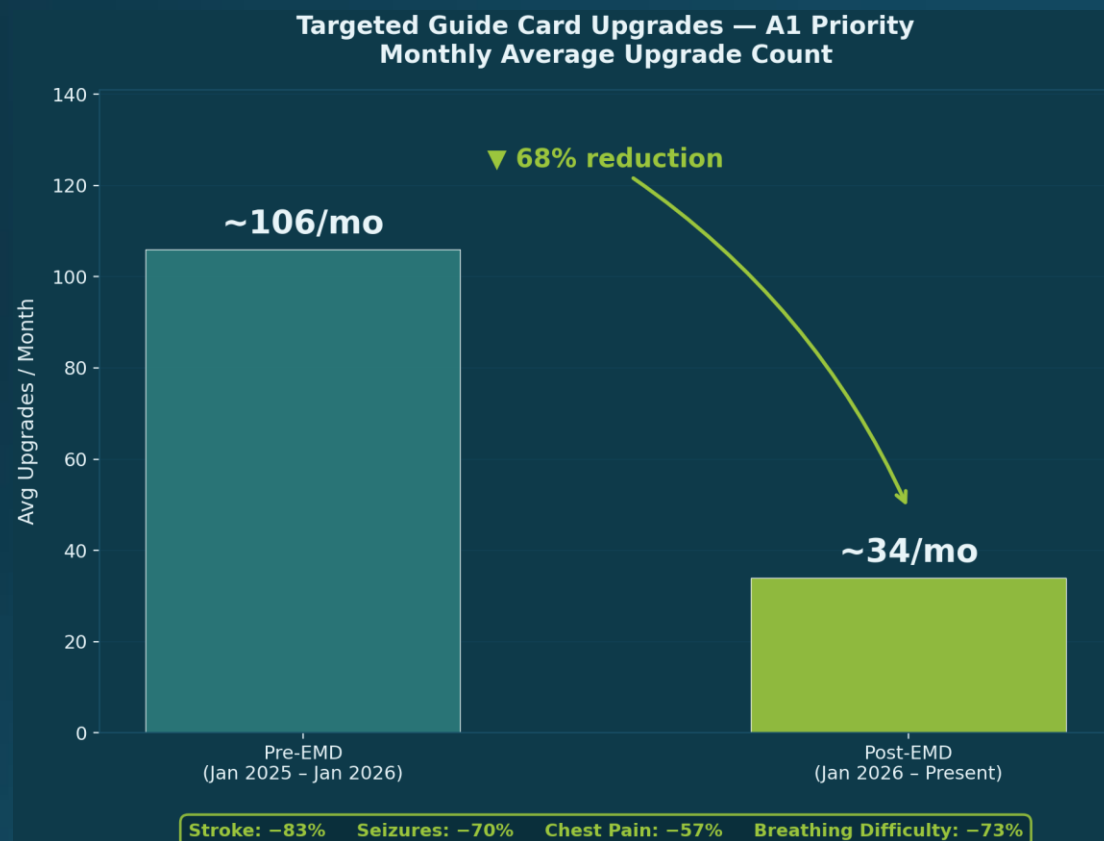
- System is more balanced than before
- Approximately 74.2% of EMS calls depend on BLS Transport as priority if available
- Increase in ALS Transports helped balance. But did not help with response times

A1 Upgrades Decreased by 52%

Monthly Upgrade Count — A1
Pre-EMD vs Post-EMD | Dashed = Pre-EMD full-month average (302 upg/mo)



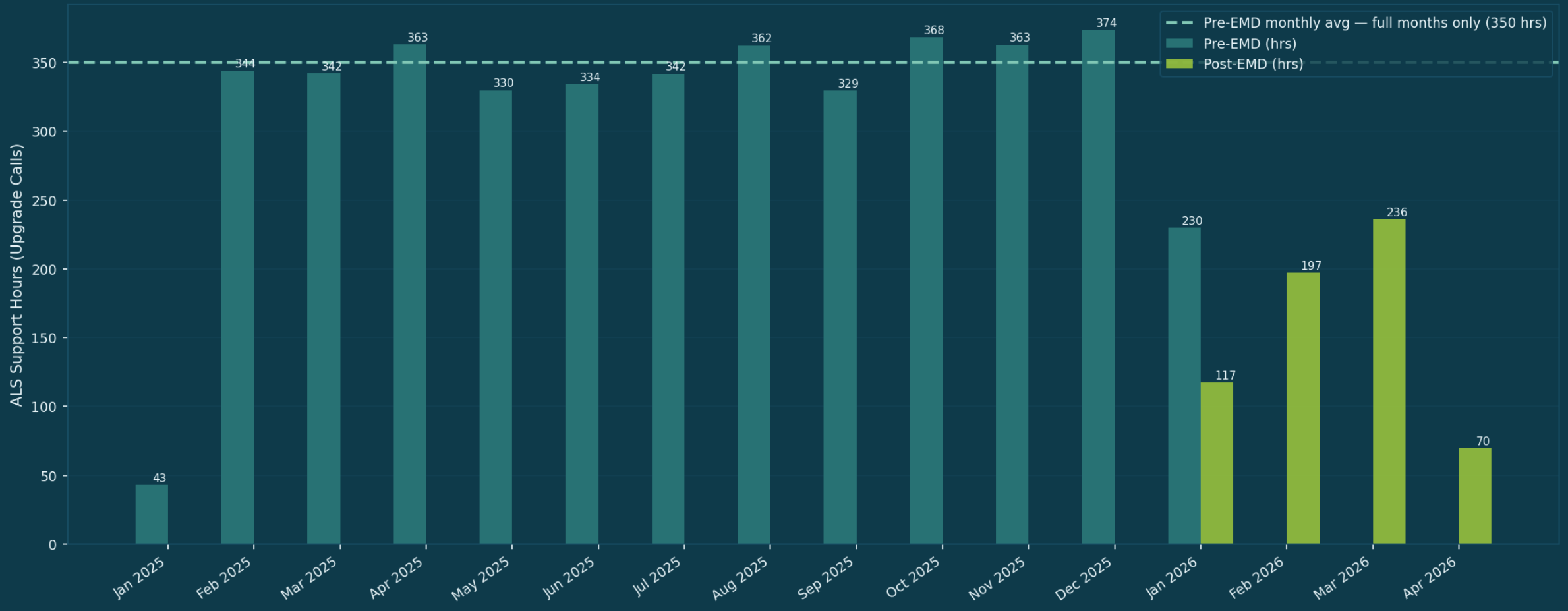
Additional Results



- 68% fewer upgrades across the 4 highest-impact guide cards
- Overall rate for A1 calls dropped from 10.5/day to 5.5/day
- On pace for over 1700 fewer trips to the hospital by an ALS Support Piece over the next year

Saving ~ 2.5 Hours of ALS Support Time Per Day

Monthly ALS Support Burden — Overall Upgrades
Pre-EMD vs Post-EMD | Dashed = Pre-EMD full-month average



Future Research

- Behavioral Study
- Disparity in Refusals
- Trauma vs. Medical
- Post - Implementation

Discussion

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